Cardiovascular System (CARD) · NPLEX-P1-CARD-08

CV-8 Coronary Circulation

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-08-01-C01				
NPLEX-P1-CARD-08-01-C02				
NPLEX-P1-CARD-08-01-C03				
NPLEX-P1-CARD-08-01-C04				
NPLEX-P1-CARD-08-01-C05				
NPLEX-P1-CARD-08-01-C06				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-08-01-C01-BD1

Draw the coronary arteries and their territories

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-8 Coronary Circulation Competency: NPLEX-P1-CARD-08-01-C01 Activity: NPLEX-P1-CARD-08-01-C01-BD1 Date: _____

Without looking at your notes, draw the heart from the front and from below. Mark the coronary sulcus, the anterior and posterior interventricular grooves, and the crux. Draw both coronary arteries from their origins in the aortic sinuses, name every branch in order along each artery, and write beside each branch the muscle and the part of the septum it supplies. Give the approximate percentage of hearts in which the right coronary artery gives the sinoatrial nodal artery and the atrioventricular nodal artery. Shade the territory of each artery on your drawing and write the electrocardiogram lead group that looks at each territory. Finish by writing which artery is occluded most often in infarction and why that follows from how much muscle it feeds.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-08-01-C01-BD2

Work backward from the tracing to the artery

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-8 Coronary Circulation Competency: NPLEX-P1-CARD-08-01-C01 Activity: NPLEX-P1-CARD-08-01-C01-BD2 Date: _____

Three patients each have an acute coronary occlusion. Patient 1 has ST elevation in V1 to V4, patient 2 in II, III, and aVF, and patient 3 in I, aVL, V5, and V6. For each patient name the occluded artery, the wall and the part of the septum that is losing its supply, and the conduction structures at risk, then write the conduction abnormality you expect and whether the block sits at the atrioventricular node or below it. Next, write why a stenosis that developed over two years can leave the muscle beyond it alive while a sudden thrombus in the same artery infarcts it, and name the two places in the heart where the anastomoses that allow this are best developed. Finish by writing why the right ventricle is damaged in some right coronary occlusions but almost never in left anterior descending occlusions.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-08-01-C02-BD1

Dominance and the supply of the papillary muscles

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-8 Coronary Circulation Competency: NPLEX-P1-CARD-08-01-C02 Activity: NPLEX-P1-CARD-08-01-C02-BD1 Date: _____

Write the definition of coronary dominance in one sentence, naming the single artery that the definition turns on. Then build a short table with one row for right dominance, left dominance, and codominance, giving the source of the posterior descending artery and the approximate frequency of each pattern. Beside the table write the three structures the posterior descending artery supplies and the other nodal branch that usually comes from the same artery. Underneath, write what the papillary muscles do during systole, then write for each papillary muscle which arteries supply it and whether that supply is single or dual.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-08-01-C02-BD2

Sudden decompensation four days after an inferior infarct

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-8 Coronary Circulation Competency: NPLEX-P1-CARD-08-01-C02 Activity: NPLEX-P1-CARD-08-01-C02-BD2 Date: _____

A patient had an inferior infarct four days ago and now becomes acutely breathless and hypotensive with a new systolic murmur. Write which structure has most likely failed, which artery supplied it, and why this complication appears on day four rather than on day one, naming the cells that act on the dead muscle and what they do to its strength. Then explain, step by step from pressures and volumes, why pulmonary edema comes on this abruptly and why forward stroke volume falls, and write why the murmur can be short and soft even though the regurgitation is severe. Next, write the other post-infarct rupture that causes sudden decompensation with a new murmur, and name the one measurement that separates the two. Finish by writing how the same event would present in a patient whose heart is left dominant, and what you would expect on the tracing in that patient.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-08-01-C02-RT01

Rebuild the chain from inferior infarct to mitral regurgitation

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-8 Coronary Circulation Competency: NPLEX-P1-CARD-08-01-C02 Activity: NPLEX-P1-CARD-08-01-C02-RT01 Date: _____

Rebuild the seven-step chain that runs from a thrombus in the artery giving the posterior descending artery to acute severe mitral regurgitation, one step per line and in order. Beside the step where the muscle loses its supply, write which papillary muscle it is and why only that one is at risk. Beside the softening step, write the days after infarction when it happens and the cells responsible. Beside the final step, write what happens to left atrial pressure and to forward stroke volume. Then write the one change in the chain if the heart is left dominant.

Rebuild from memory: From inferior infarct to papillary muscle rupture

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-08-01-C02-CP01

Anterolateral against posteromedial papillary muscle

Fill in the table from memory, then check it against the notes and correct any cell you missed.

Feature	Anterolateral papillary muscle	Posteromedial papillary muscle
Wall of the left ventricle it sits on		
Arteries that supply it, and whether the supply is single or dual		
Infarct location that puts it at risk		
Relative frequency of rupture		
Effect of a left anterior descending occlusion on its perfusion		
What changes about its supply in a left dominant heart		
Consequence for the mitral valve if it ruptures		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-08-01-C02-CA01

Breathlessness and a new murmur five days after chest pain

A 64-year-old man is readmitted five days after an inferior myocardial infarction. He woke an hour ago with severe breathlessness and cannot lie flat. Temperature is 36.8 °C (98.2 °F), pulse is 118/min, respirations are 32/min, and blood pressure is 84/58 mm Hg. He has coarse crackles throughout both lung fields and a new short systolic murmur at the apex that is soft at the left sternal border. The electrocardiogram taken at the first admission showed ST elevation in leads II, III, and aVF with a brief period of a ventricular rate of 44/min and P waves at 80/min. Oxygen saturation measured in the right atrium is 62% and in the right ventricle is 63%.

1. Recognize. Name the structure that has failed, the artery that supplied it, and the valve lesion that has resulted.

2. Explain. Explain from the blood supply of that structure why it was the one at risk in this infarct, and explain why the failure happened on day five rather than during the first day.

3. Differentiate. Write how this presentation differs from post-infarct ventricular septal rupture and from ischemic papillary muscle dysfunction without rupture, naming the feature that separates each from this patient.

4. Investigate. State what the right atrial and right ventricular saturations in this patient tell you, and write what they would have shown in the lesion you ruled out.

5. Decide. State which layer and which wall of the left ventricle became ischemic first in the original infarct, and name the conduction structure that explains the ventricular rate of 44/min on the first tracing.

6. Justify. Justify the abrupt pulmonary edema and the hypotension from the pressures and volumes involved, and explain why the murmur is short and soft despite severe regurgitation.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-08-01-C03-BD1

Build the cardiac venous drainage table from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-8 Coronary Circulation Competency: NPLEX-P1-CARD-08-01-C03 Activity: NPLEX-P1-CARD-08-01-C03-BD1 Date: _____

On blank paper build a table of cardiac veins with one row each for the great cardiac vein, the middle cardiac vein, the small cardiac vein, the anterior cardiac veins, and the smallest cardiac veins. Use the columns course and companion artery, region drained, and termination, and fill every cell. Underneath, write where the coronary sinus lies, where it opens into the right atrium relative to the inferior vena cava and the tricuspid orifice, the structure that partly guards its opening, and the triangle whose border its ostium forms. Then write the approximate fraction of total coronary flow that returns through the sinus and which ventricle contributes most of it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-08-01-C03-BD2

Read the venous side as a measurement

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-8 Coronary Circulation Competency: NPLEX-P1-CARD-08-01-C03 Activity: NPLEX-P1-CARD-08-01-C03-BD2 Date: _____

Blood sampled from the coronary sinus of a healthy adult is about 25% to 30% saturated while mixed venous blood is about 75% saturated. Write why the coronary figure is so much lower, give the oxygen content of arterial and coronary sinus blood in mL O2/dL, and write what the coronary sinus sample tells you about left ventricular metabolism. Next, an echocardiogram in another patient shows a dilated coronary sinus. Write the embryological structure whose persistence explains this, where the sinus itself comes from in development, and why the finding is usually harmless. Finish by writing which two sources of venous blood make up the normal physiological shunt of about 2% of cardiac output, and explain in two sentences why that shunt keeps the alveolar to arterial oxygen difference above zero in a healthy person.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-08-01-C04-BD1

Draw coronary flow across the cardiac cycle

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-8 Coronary Circulation Competency: NPLEX-P1-CARD-08-01-C04 Activity: NPLEX-P1-CARD-08-01-C04-BD1 Date: _____

Draw one cardiac cycle with aortic pressure, left ventricular pressure, and left coronary flow on the same time axis. Mark where coronary flow falls, where it surges, and where it is highest, and write beside each mark what is happening to the vessels inside the wall. Write the approximate percentage of left ventricular flow that occurs in diastole, the resting coronary flow in mL/min, and that flow as a percentage of cardiac output alongside the heart as a percentage of body weight. Write the equation for left ventricular coronary perfusion pressure and work out a normal value from a blood pressure of 120/80 mm Hg and a left ventricular end-diastolic pressure of 8 mm Hg. Then write why the right ventricle is perfused throughout the cycle instead, and the one condition that changes that. Finish with the oxygen content of arterial and coronary venous blood, the resting extraction percentage for the heart and for the whole body, and the four features of the myocyte that force extraction to be this high.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-08-01-C04-BD2

Four patients with ischemia and normal arteries

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-8 Coronary Circulation Competency: NPLEX-P1-CARD-08-01-C04 Activity: NPLEX-P1-CARD-08-01-C04-BD2 Date: _____

Four patients have chest pain on exertion with no coronary stenosis: one has severe aortic stenosis, one has severe aortic regurgitation with a blood pressure of 150/40 mm Hg and a left ventricular end-diastolic pressure of 25 mm Hg, one has a sustained heart rate of 160/min, and one has a hemoglobin of 6 g/dL (reference range 12 to 16 g/dL). For each patient calculate or state what happens to coronary perfusion pressure, to diastolic perfusion time per minute, to wall stress, and to oxygen demand, and write which term in the supply and demand balance has failed. Write the Laplace relationship you are using and show the aortic regurgitation calculation. Then write which layer of the left ventricle becomes ischemic first in all four patients and give the four reasons that layer loses out, and write why hypertrophy lowers wall stress and still leaves the inner wall vulnerable. Finish by writing what the electrocardiogram shows in these patients during exertion and why that pattern rather than ST elevation.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-08-01-C04-RT01

Rebuild the tachycardia sequence with the arithmetic

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-8 Coronary Circulation Competency: NPLEX-P1-CARD-08-01-C04 Activity: NPLEX-P1-CARD-08-01-C04-RT01 Date: _____

Rebuild the six-step sequence that runs from a rising heart rate to inner wall ischemia, one step per line and in order. Beside the step about shortening diastole, work the arithmetic for 70/min and for 150/min: give the cycle length, the duration of systole, the duration of diastole, and the total diastolic seconds per minute at each rate, then give the percentage fall in perfusion time. Beside the step about demand, write why each extra beat costs oxygen. Beside the last step, write the layer involved and what appears on the tracing. Finish by writing the two things happening at once that make tachycardia so poorly tolerated when a stenosis is present.

Rebuild from memory: How tachycardia produces subendocardial ischemia

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-08-01-C04-PR01

Predict the effect of a fast atrial rhythm in aortic stenosis

A patient with severe aortic stenosis and a hypertrophied, stiff left ventricle develops a sustained ventricular rate of 150/min. Predict each item below, write the direction of the change, and give the mechanism in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to the duration of diastole in each cycle and to total diastolic seconds per minute		
What happens to left ventricular end-diastolic pressure in this stiff ventricle		
What happens to coronary perfusion pressure once end-diastolic pressure has changed		
What happens to myocardial oxygen demand, using cavity pressure, radius, wall thickness, and rate		
What happens to the vasodilator reserve still available in the subendocardium		
Which layer of the wall becomes ischemic first and what appears on the electrocardiogram		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-08-01-C05-BD1

Rebuild metabolic control of coronary flow

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-8 Coronary Circulation Competency: NPLEX-P1-CARD-08-01-C05 Activity: NPLEX-P1-CARD-08-01-C05-BD1 Date: _____

Write the Fick principle as it applies to myocardial oxygen consumption, then explain in three sentences why the heart cannot meet higher demand the way skeletal muscle does, using the resting and maximal extraction percentages. Next, rebuild the adenosine pathway in seven steps from an oxygen deficit to a rise in flow, naming the enzyme that makes adenosine, the receptor and G protein it acts through, the second messenger and kinase, the channels that open, the effect on membrane voltage, the channel that then closes, the ion whose cytosolic level falls, and the enzyme whose activity drops before the muscle relaxes. Write how adenosine is cleared and how fast. Then list every other mediator that contributes in parallel, and write what sympathetic alpha-1 and beta-1 activity each do to coronary flow and which effect wins. Finish by writing the perfusion pressure range over which flow is autoregulated and the definition and normal value of coronary flow reserve.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-08-01-C05-BD2

Work the numbers on a stenosis, then explain steal

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-8 Coronary Circulation Competency: NPLEX-P1-CARD-08-01-C05 Activity: NPLEX-P1-CARD-08-01-C05-BD2 Date: _____

Take resting coronary flow as 2.25 dL/min, arterial oxygen content as 20 mL/dL, and coronary venous content as 6 mL/dL. Calculate the arteriovenous difference, resting myocardial oxygen consumption, and resting extraction, showing your working. Now demand doubles. Calculate the consumption that maximal extraction alone would allow at unchanged flow, then calculate the flow needed to meet the new demand at the resting extraction, and write one sentence on what this proves about a lesion that caps flow. Next, a patient has a 75% diameter stenosis of the left anterior descending artery. Write what resting flow, maximal flow, and arteriolar tone are doing in that territory, give the degree of narrowing at which maximal flow begins to fall and the degree at which resting flow begins to fall, and explain why the patient is comfortable sitting and has chest pain climbing stairs. Finish by writing what happens to flow in the stenosed bed and in the normal beds when a vasodilator is given, name the two drugs the notes use as examples and how the second one raises adenosine levels, and write why this effect is useful during stress perfusion imaging.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-08-01-C06-BD1

Rebuild why ALCAPA fails only after birth

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-8 Coronary Circulation Competency: NPLEX-P1-CARD-08-01-C06 Activity: NPLEX-P1-CARD-08-01-C06-BD1 Date: _____

Write what is anatomically abnormal in anomalous origin of the left coronary artery from the pulmonary artery, and which coronary artery arises normally. Then rebuild in six steps the sequence from the fetal circulation to infant myocardial ischemia, one step per line. Beside the fetal steps, write what keeps pulmonary artery pressure about equal to aortic pressure, the direction of flow in the anomalous artery, and the two reasons the fetal heart tolerates the lesion. Beside the postnatal steps, write the pulmonary artery pressure and the oxygen saturation the left coronary artery now receives, the time course over which this develops, and why collateral flow from the right coronary artery ends up draining into the pulmonary trunk rather than perfusing muscle. Name that last phenomenon. Finish by drawing the self-reinforcing cycle in six steps, from the ischemic ventricle through dilation, wall stress, mitral regurgitation, and end-diastolic pressure, and write beside each step whether it raises demand or lowers supply.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-08-01-C06-BD2

The infant who looks well at birth and sick at ten weeks

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-8 Coronary Circulation Competency: NPLEX-P1-CARD-08-01-C06 Activity: NPLEX-P1-CARD-08-01-C06-BD2 Date: _____

A 10-week-old infant is pale, sweaty, and breathless during feeds and has gained little weight. Write why symptoms begin at this age rather than in the first week, and why feeding is the exertion that brings them on. List the findings you expect on examination and on the chest film, then write the electrocardiogram pattern with the exact leads involved and why that finding in an infant is so unusual, and write the two mechanisms that make this infant regurgitate through the mitral valve. Next, write the two features that separate this diagnosis from dilated cardiomyopathy and myocarditis. Then write why a minority of these patients instead reach childhood or adulthood, listing what you expect to find in them and the one saturation measurement that can be slightly unexpected and why. Finish by writing why a large ventricular septal defect in the same infant can delay or hide the ischemia, and state in one sentence the variable that actually decides whether the left ventricle is perfused.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-08-01-C01-Q01

Locating the occlusion from the complication

A 59-year-old man comes to the emergency department because of 2 hours of chest pain. Pulse is 48/min and blood pressure is 92/60 mm Hg. The electrocardiogram shows ST elevation in leads II, III, and aVF, with P waves at 82/min and QRS complexes at 46/min of 0.08 s duration that are unrelated to the P waves. Which of the following arteries was most likely occluded, and which structure accounts for the slow ventricular rate?

A. Left anterior descending artery supplying the bundle branches

B. Circumflex artery supplying the lateral wall

C. Right coronary artery supplying the atrioventricular node

D. Left main coronary artery supplying most of the left ventricle

Why did you pick it?

Board-style question NPLEX-P1-CARD-08-01-C04-Q02

Ischemia with unobstructed arteries

A 71-year-old woman with severe aortic regurgitation has chest pain on exertion. Blood pressure is 160/40 mm Hg and left ventricular end-diastolic pressure measured at catheterization is 26 mm Hg. Coronary angiography shows no stenosis. Which of the following best explains her exertional chest pain?

A. A coronary perfusion pressure of about 14 mm Hg with high wall stress in a dilated ventricle, which exhausts subendocardial reserve

B. Failure of myocardial oxygen extraction to rise above 25% during exertion

C. Loss of systolic coronary flow to the subepicardium of the left ventricle

D. Diversion of flow from the left coronary artery into the pulmonary trunk

Why did you pick it?

Board-style question NPLEX-P1-CARD-08-01-C06-Q03

Why the infant deteriorates at two months

A 9-week-old infant has episodes of pallor, sweating, and irritability during feeds, with poor weight gain. The chest film shows cardiomegaly and the electrocardiogram shows abnormal Q waves with ST and T changes in leads I, aVL, V5, and V6. Echocardiography shows that the left coronary artery arises from the pulmonary trunk. Which of the following best explains why the infant was well at birth and is ischemic now?

A. Progressive atherosclerotic narrowing of the anomalous left coronary artery

B. Closure of the foramen ovale, which removed a right to left shunt that had been perfusing the left ventricle

C. A rise in myocardial oxygen extraction above the fetal level, which exceeded the capacity of the anomalous artery

D. A fall in pulmonary vascular resistance and pulmonary artery pressure, which left the anomalous artery with low-pressure desaturated blood and reversed collateral flow into the pulmonary trunk

Why did you pick it?